

UNIVERSITY EXAMINATION BOARD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MIDTERM EXAMINATION (BACKTRACKING OPTIMIZED)

Subject: Advanced Data Structures and Algorithms	Sub Code: CS-501	Max Marks: 50
Time / Duration: 3 Hours	Academic Year: 2026-2027	Semester: V Semester

General Instructions:

- 1. Answer all questions.
- 2. Assume suitable data if necessary.

PART A (COMPULSORY CONCEPTUAL BITS)

SHORT QUESTIONS

1. Define Divide and conquer design paradigms and state its significance in Advanced Data Structures and Algorithms. **[2 Marks]**
2. Define Greedy choice property and state its significance in Advanced Data Structures and Algorithms. **[2 Marks]**
3. Define Elements of dynamic programming and state its significance in Advanced Data Structures and Algorithms. **[2 Marks]**
4. Define Backtracking paradigm and state its significance in Advanced Data Structures and Algorithms. **[2 Marks]**
5. Define NP-Hard and state its significance in Advanced Data Structures and Algorithms. **[2 Marks]**

PART B (ANALYTICAL & DESIGN PROBLEMS)

UNIT-WISE COMPREHENSIVE QUESTIONS

6. (a) Design a robust solution using Divide and conquer design paradigms for an enterprise-scale engineering problem. Analyze its complexity. **[10 Marks]**

[OR]

- (b) What is the primary function or objective of Divide and conquer design paradigms? **[10 Marks]**
- 7. (a) Critically evaluate Greedy choice property against alternative paradigms. When is Greedy choice property strictly optimal? **[10 Marks]**

[OR]

- (b) What is the primary function or objective of Greedy choice property? **[10 Marks]**
- 8. (a) Derive and explain the mathematical recurrence or governing equations of Elements of dynamic programming. Prove its correctness. **[10 Marks]**

[OR]

- (b) What is the primary function or objective of Elements of dynamic programming? **[10 Marks]**

9. (a) Critically evaluate Backtracking paradigm against alternative paradigms. When is Backtracking paradigm strictly optimal? **[10 Marks]**

[OR]

(b) What is the primary function or objective of Backtracking paradigm? **[10 Marks]**